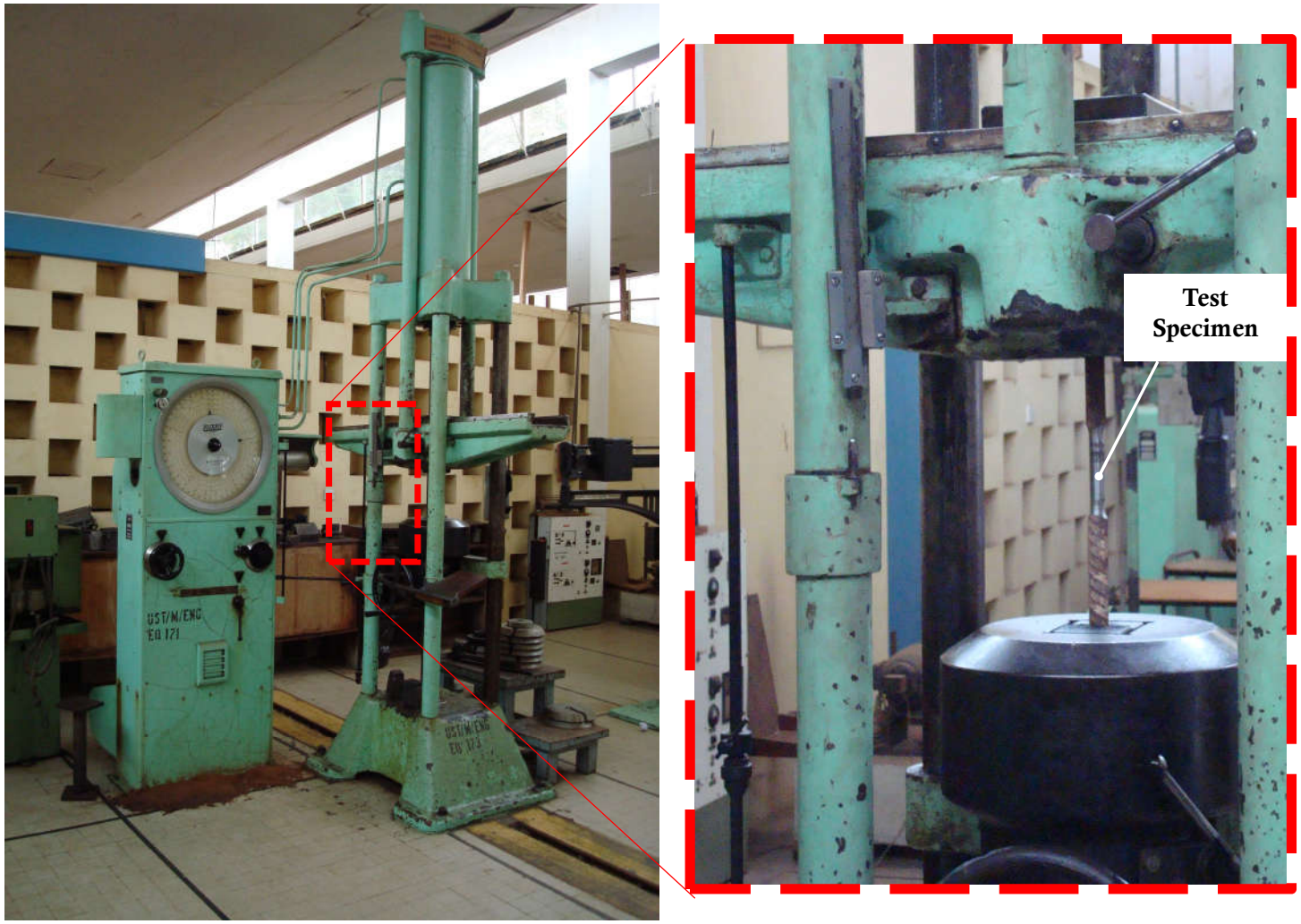
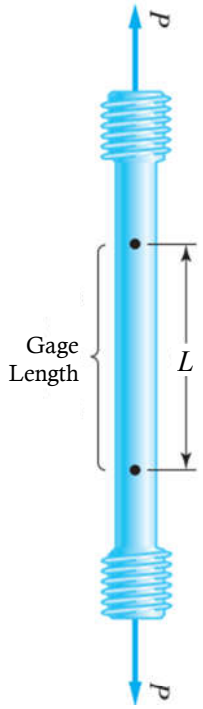
Stress-Strain Diagrams/Relationships (Mechanical Behavior of Materials)

Tensile Tests

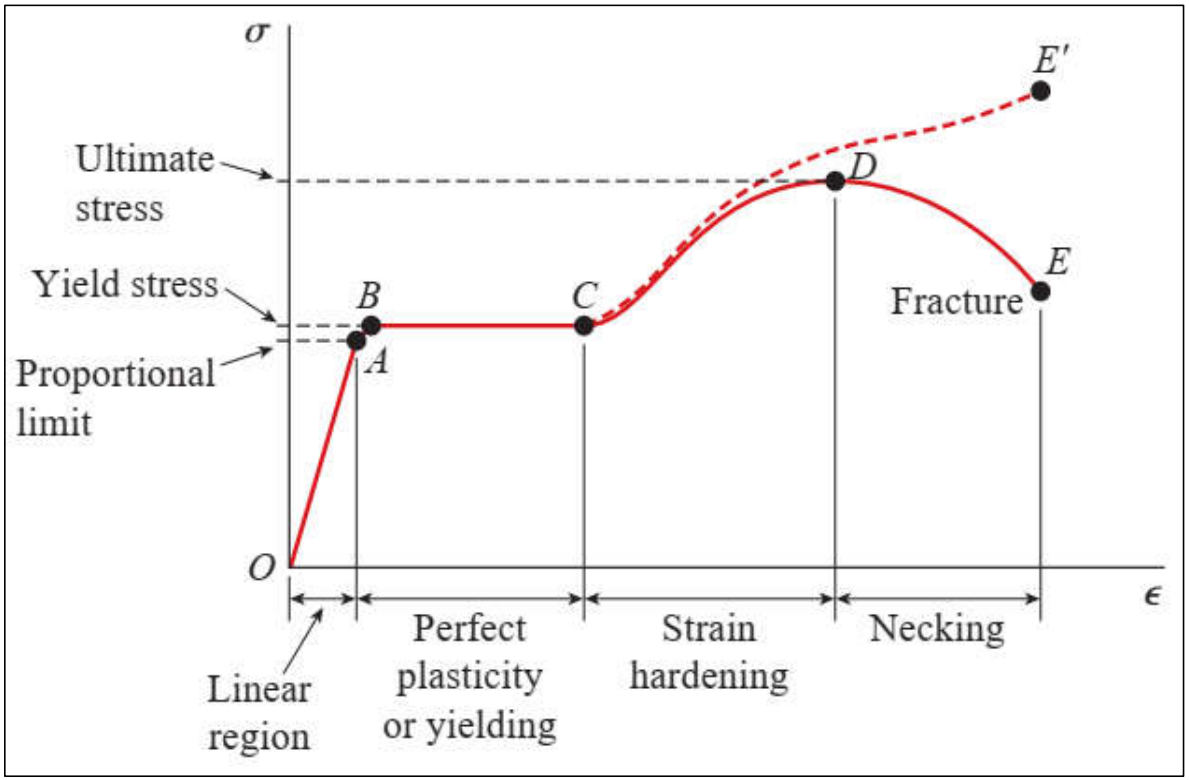


Test Specimen are prepared according to standards.

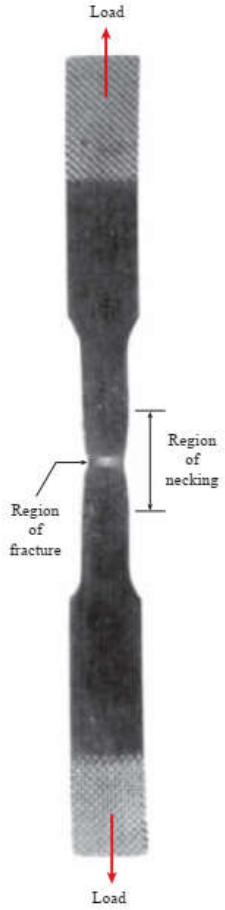
Eg. American Society for Testing and Materials (ASTM)



Stress-Strain Diagrams



True Stress - - - - Engineering Stress ——



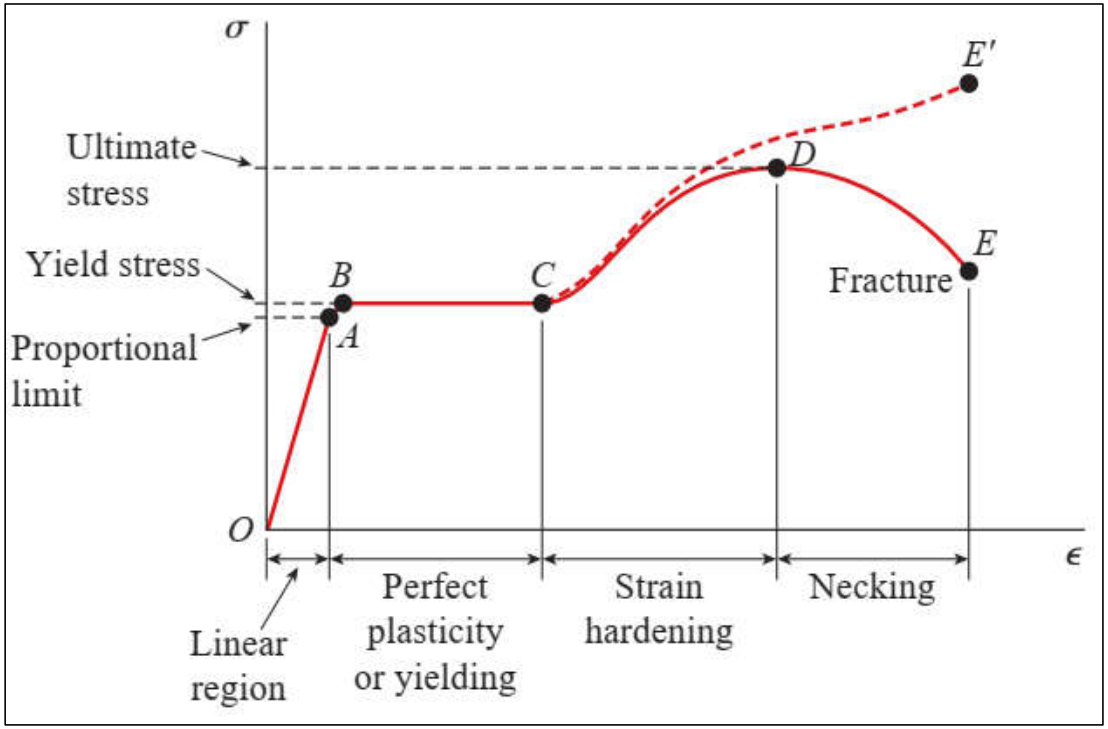


Fracture in a Ductile material



Fracture in a Brittle material

Some Mechanical Properties of Materials



Strength:
Capacity for high stress/ultimate stress.

Toughness:
Capacity for energy absorption (area under stress strain curve)

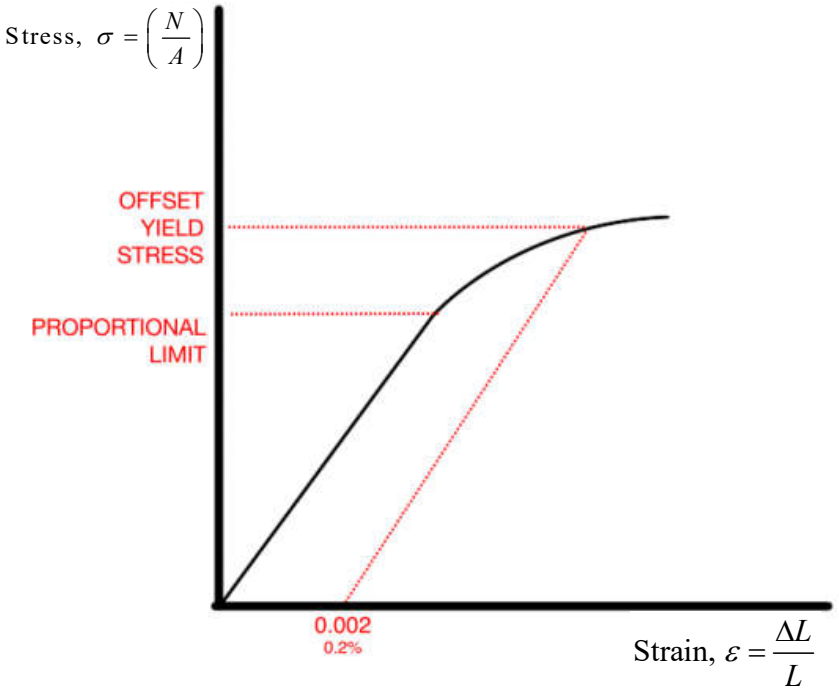
Resilience:
capacity for elastic deformation (area under elastic region)

Ductility:
high capacity for deformation/strain

Brittleness:
low capacity for deformation/strain

Stiffness:
the slope in the Linear Elastic Region. A stiffer material will have a steeper slope.

Some Stress-Strain Relationships



- 0.2% Offset Rule
  - Where the yield stress is not clearly defined, the 0.2% offset yield stress can be used.

- Hooke's Law:

  For simple tension and compression in the linear elastic region,

  $\sigma = E\varepsilon$

  E is the Modulus of Elasticity/Young's Modulus.

- Change in length (axial deformation) relation:

  $$\Delta L = \delta = \frac{NL}{AE}$$

- For shear, Hooke's Law can be stated as

  $\tau = G\gamma$

  G is the Modulus of Rigidity/Shear Modulus.

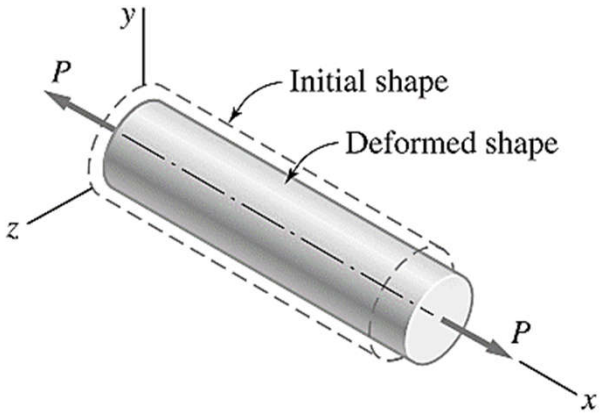
- The moduli of Elasticity and Rigidity are related via the Poisson's Ratio (ν) as

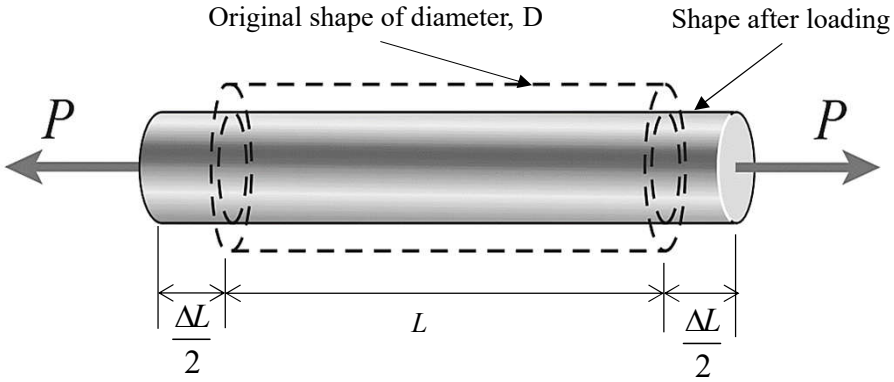
  $E = 2(1+\nu)G$

Stress-Strain Relationships

- Poisson's Ratio

After uniaxial loading of a member,





Longitudinal Strain, $\varepsilon_x = \dfrac{\delta_{longitudinal}}{L} = \dfrac{\Delta L}{L}$

Lateral Strain, $\varepsilon_y = -\dfrac{\delta_{lateral}}{d} = -\dfrac{\Delta d}{d}$

Poisson's Ratio, $\nu = \dfrac{\text{Lateral Strain}}{\text{Longitudinal Strain}} = -\dfrac{\varepsilon_y}{\varepsilon_x} = -\dfrac{\varepsilon_z}{\varepsilon_x}$

And $\varepsilon_x = \dfrac{\sigma_x}{E}$

For an isotropic material, $-\nu\varepsilon_x = \varepsilon_y = \varepsilon_z$

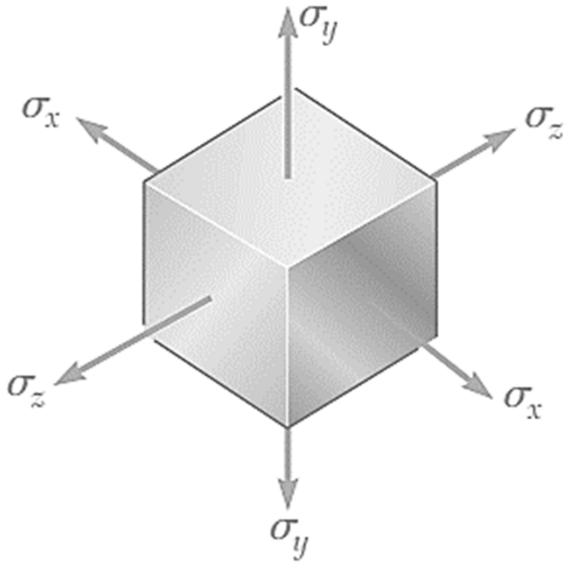
$$\Rightarrow \varepsilon_y = \varepsilon_z = -\nu\frac{\sigma_x}{E}$$

Note: Poisson's ratio remains constant only in the linear elastic range.

Stress-Strain Relationships

- Poisson's Ratio

  For multiaxial loading



$$\varepsilon_x = +\frac{\sigma_x}{E} - \nu\frac{\sigma_y}{E} - \nu\frac{\sigma_z}{E}$$

$$\varepsilon_y = -\nu\frac{\sigma_x}{E} + \frac{\sigma_y}{E} - \nu\frac{\sigma_z}{E}$$

$$\varepsilon_z = -\nu\frac{\sigma_x}{E} - \nu\frac{\sigma_y}{E} + \frac{\sigma_z}{E}$$

Factors of Safety

- To ensure safety applied stresses on structures are restricted to values less than they can fully support.
- The failure (yield or ultimate) stress of the material a member is to be made of, is divided by a factor of safety to determine allowable stresses it can withstand when in use.

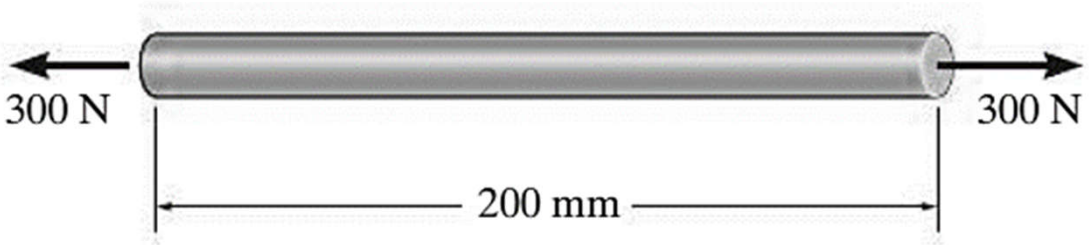
$$\text{Factor of Safety, } FoS = \frac{\sigma_{failure\ of\ material}}{\sigma_{\max}} = \frac{Strength\ of\ Material\ (Yield\ Stress,\ Ultimate\ Stress)}{Maximum\ Computed\ Stress}$$

- FoS > 1 is recommended.
- FoS values depend several factors; type of material (higher in brittle materials - ultimate strength preferred), environment in which part will be used, type of application.
- Typically values (Mechanics of Materials, W. Whiteman);
  - Buildings; FoS $\geq$ 2
  - Automobiles; FoS $\geq$ 3
  - Boilers/Pressure vessels/Bolts; FoS $\geq$ 8.5
  - Aircraft/Spacecraft; FoS $\geq$ 1.2-2.5
  - Lifting Equipment/Hooks; FoS $\geq$ 8-9

Stress-Strain Relationships

Example

The plastic rod shown is 200-mm long and 15-mm in diameter. If an axial load of 300-N is applied to it, determine the change in its length and the change in its diameter. Take E = 2.70-GPa and $\nu = 0.4$



$$\sigma = \frac{N}{A} = \frac{300}{\pi(0.015)^2/4} = 1.678 \text{ MPa}$$

$$\varepsilon_{long} = \frac{\sigma}{E} = 0.0006288$$

$$\delta_{long} = \Delta L = \varepsilon_{long} L = 0.126 \text{ mm}$$

$$\frac{\delta_{dia}}{d} = \frac{\Delta d}{d} = \varepsilon_{lat}$$

$$\varepsilon_{lat} = -\nu\varepsilon_{long} = -0.0002515$$

$$\delta_{dia} = \Delta d = \varepsilon_{lat} d = -0.00377 \text{ mm}$$

Alternatively,

From load-displacement relation,

$$\delta_{long} = \Delta L = \frac{NL}{AE} = \frac{300(0.2)(4)}{\pi(0.015^2)(2.7\times10^9)} = 0.000126 \text{ m}$$

$$\varepsilon_{long} = \frac{\Delta L}{L} = \frac{0.126}{200} = 0.00063$$

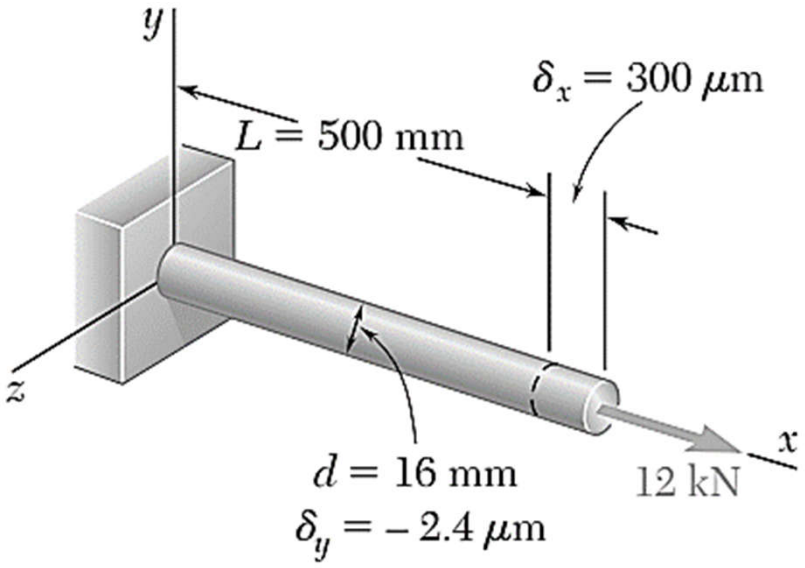
$$\frac{\delta_{dia}}{d} = \frac{\Delta d}{d} = \varepsilon_{lat} = -\nu\varepsilon_{long}$$

$$\delta_{dia} = \Delta d = -0.00377 \text{ mm}$$

Stress-Strain Relationships

Example

A 500-mm long, 16-mm diameter rod made of a homogenous, isotropic material is observed to increase in length by 300-μm, and to decrease in diameter by 2.4-μm when subjected to an axial 12-kN load. Determine the modulus of elasticity and Poisson's ratio of the material. Assume the material remains elastic.



$$\sigma_x = \frac{N}{A} = \frac{12000}{\pi(0.016)^2/4} = 59.7 \text{ MPa}$$

$$\varepsilon_x = \frac{\delta_x}{L} = \frac{\Delta L}{L} = 0.0006$$

$$\varepsilon_y = \frac{\delta_{dia}}{d} = \frac{\Delta d}{d} = -0.00015$$

$$E = \frac{\sigma_x}{\varepsilon_x} = 99.5 \text{ GPa}$$

$$\nu = -\frac{\varepsilon_y}{\varepsilon_x} = 0.25$$

Alternatively, From load-displacement relation,

$$E = \frac{NL}{A\delta_x} = \frac{12000(0.5)(4)}{\pi(0.016^2)(300\times10^{-6})} = 99.5 \text{ GPa}$$

$$\nu = -\frac{\varepsilon_{lat}}{\varepsilon_{long}} = -\frac{\varepsilon_y}{\varepsilon_x}$$

$$\varepsilon_{long} = \varepsilon_x = \frac{\delta_x}{L} = \frac{\Delta L}{L} = 0.0006$$

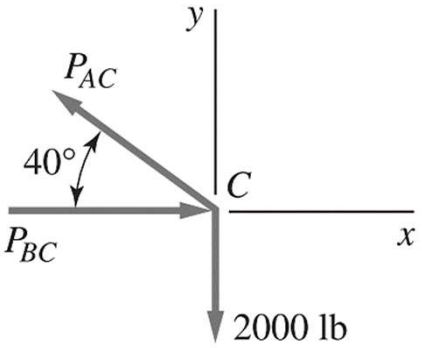
$$\varepsilon_{lat} = \varepsilon_y = \frac{\delta_d}{d} = \frac{\Delta d}{d} = -0.00015$$

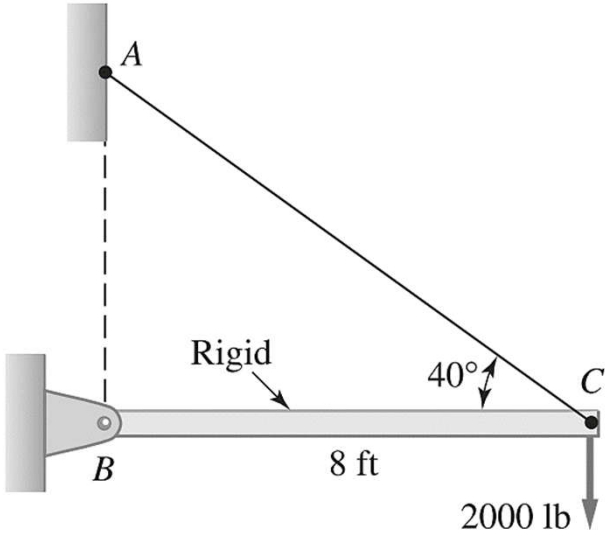
$$\nu = 0.25$$

Axial (Elastic) deformation

Example

The rigid bar BC is supported by the steel rod AC of cross-sectional area 0.25 in^2 as shown below. Find the extension in rod AC caused by the 2000-lb load. Use E = 29 x 10^6 psi for steel.





$$P_{AC} = \frac{2000 \text{ lb}}{\sin 40} = 3111.45 \text{ lb}$$

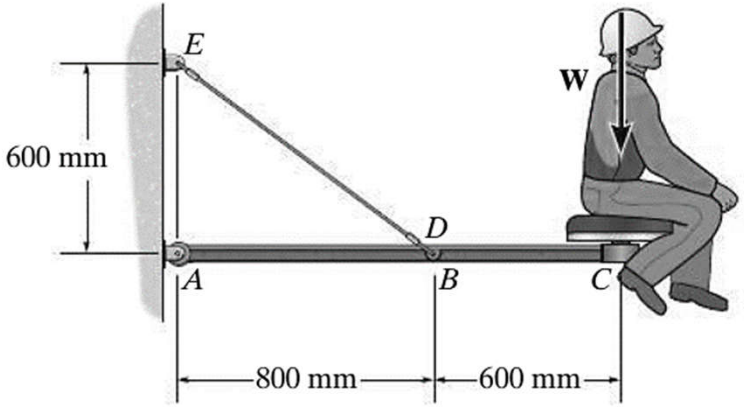
$$L_{AC} = \frac{8 \text{ ft}}{\cos 40} \times 12 = 125.32 \text{ in}$$

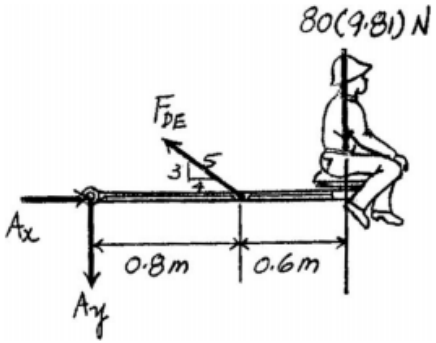
$$\delta_{AC} = \Delta L_{AC} = \frac{NL}{AE} = \frac{P_{AC} L_{AC}}{A_{AC} E_{AC}} = 0.05378 \text{ in}$$

Axial deformation

Example

The wire has a diameter of 5 mm and is made from A-36 steel. If a 80-kg man is sitting on seat C, determine the elongation of wire DE. A-36 steel has a Young's Modulus, E of 200-GPa.





$$\sum M_A = 0; \Rightarrow F_{DE} = 2289 \text{ N}$$

$$\sigma_{DE} = \frac{F_{DE}}{A_{DE}} = 116.58 \text{ MPa}$$

$$\varepsilon_{DE} = \frac{\sigma_{DE}}{E} = 0.0005829$$

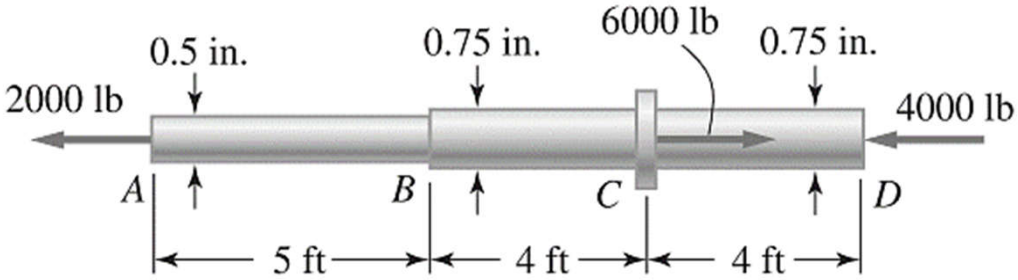
$$\delta_{DE} = \Delta L_{DE} = \varepsilon_{DE} L_{DE} = 0.583 \text{ mm}$$

OR

$$\delta_{DE} = \Delta L_{DE} = \frac{F_{DE} L_{DE}}{A_{DE} E}$$

Axial deformation

- Composite member of Uniform Cross Sections.



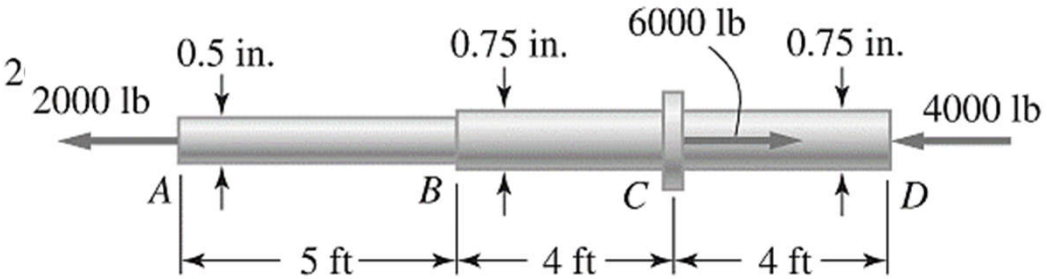
- Principle of Superposition

$$\delta_{total} = \Delta L_{total} = \sum_i \frac{N_i L_i}{A_i E_i}$$

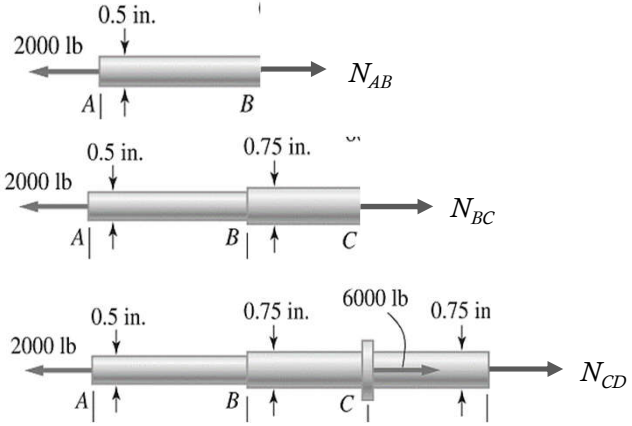
Axial deformation

Example

The steel propeller shaft ABCD carries the axial loads shown below. Determine the change in the length of the shaft caused by these loads. Use E=29 x 10^6 psi for steel.



$$\delta_{total} = \Delta L_{total} = \sum \frac{NL}{AE} = \left(\frac{NL}{AE}\right)_{AB} + \left(\frac{NL}{AE}\right)_{BC} + \left(\frac{NL}{AE}\right)_{CD}$$



$N_{AB} = 2000$ lb

$N_{BC} = 2000$ lb

$N_{CD} = -4000$ lb

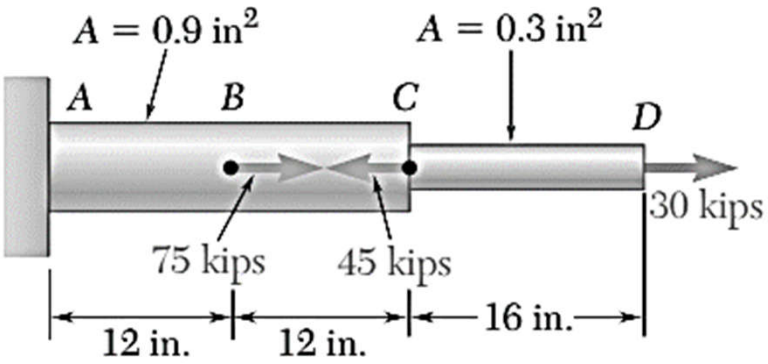
$$\therefore \delta_{total} = \frac{1}{29\times10^6}\left[\left(\frac{2000\times5\times12}{\pi(0.5)^2/4}\right)+\left(\frac{2000\times4\times12}{\pi(0.75)^2/4}\right)+\left(\frac{(-4000)\times4\times12}{\pi(0.75)^2/4}\right)\right]$$

$$= 0.01358 \text{ in}$$

Axial deformation

Example

Determine the deformation of the steel rod shown below under the given loads. Take $E_{steel} = 29 \times 10^6$ psi. 1-kip = 1000-lb



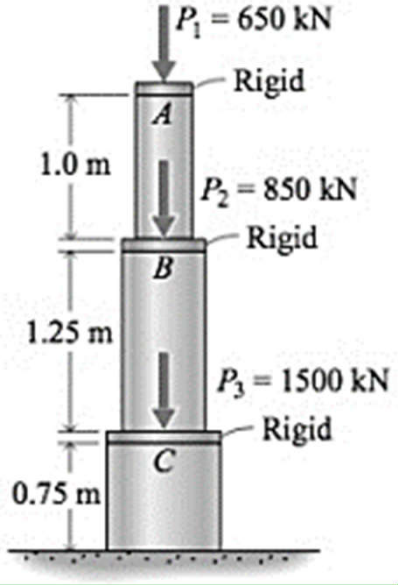
$$\delta_{total} = \sum \frac{NL}{AE} = \frac{1}{E}\left[\left(\frac{NL}{A}\right)_{AB} + \left(\frac{NL}{A}\right)_{BC} + \left(\frac{NL}{A}\right)_{CD}\right]$$

$$\delta_{total} = 75.9 \times 10^{-3} \text{ in}$$

Axial deformation

Example

The member shown comprises a solid aluminum bar A, with an outside diameter of 100-mm, a brass tube, B with outside and inside diameters of 150-mm and 100-mm respectively, and a steel pipe C with outside and inside diameters of 200-mm and 125-mm respectively. The moduli of elasticity of the aluminum, brass and steel are 73-GPa, 100-GPa, and 210-GPa respectively. Determine the overall change in length of the member under the indicated loads.



$$\delta_{total} = \sum \frac{NL}{AE} = \left(\frac{NL}{EA}\right)_{AB} + \left(\frac{NL}{EA}\right)_{BC} + \left(\frac{NL}{EA}\right)_{CD}$$

$$\delta_{total} \simeq 3.60 \text{ mm}$$